

Lecture 01: Intro

EEFE 530

Daniel Brent

Schedule

Agenda

1. Schedule
2. Mini-syllabus
3. Course Objectives
4. *MHE's* FAQ
 1. Question
 2. Experiment
 3. Identification
 4. Inference

Today

- Get to know you: what are you working on and why are you here?
- Basics of causal identification research *MHE*: Preface & Ch. 1
- What we will do in this class
- **R**: Get R installed and learn some very basic syntax
- Learn some basics of GitHub

Short run

- Get more comfortable with **R**.
- Review metrics and building intuition for causality and inference.
- Add tools to your methodological toolbox.
- Jump start your research in a structured setting.

Long run

Goal: Get comfortable with empirical causal research -
learn how to learn

GitHub for This Course

How you will use GitHub

- Access course materials (this repository)
 - Submit problem sets (GitHub Classroom)
 - Track revisions and feedback over time
-
- This is my first time using GitHub so it's a learning experience for me too.
 - Please be patient with me and I'll be patient with you.
 - Long-term it will be valuable to know. At the very least you'll get a website out of it.

Where to find things

- `README.md`: course landing page and links
- `lectures/`: lecture notes and code
- `assignments/`: instructions and templates
- `resources/`: software setup and readings

GitHub Classroom

- Should have receive a link via a Canvas announcement
- Create an account that links to your name in the roster
- Assignments should be submitted there
- You'll create repositories for each assignment

GitHub Workflow

1. Create a GitHub account (if you do not already have one)
2. Install R and RStudio
3. Install Quarto (one time)
4. Clone the course repository (or download a ZIP if you are not ready for Git yet)
5. Pull updates regularly

Rule of thumb: If something on the course page changes, you want that change locally before you start working.

Problem Set 1: Create Your Own GitHub Page

You will create a simple personal page for:

- a short bio / research interests
- links to replication exercises and course work (optional)
- professional visibility
- This will also help you get comfortable with GitHub
- Having a personal webpage is helpful as a grad student

GitHub Pages

1. Create a new repository named: `YOUR-USERNAME.github.io`
2. Add a file named `index.html` (or a `README.md`)
3. Go to **Settings** → **Pages**
4. Set “Deploy from a branch” and select `main` / root
5. Your site will be available at: `https://YOUR-USERNAME.github.io`

I just did migrated my page to GitHub: **danielbrent.com**

Publish Quarto to GitHub Pages

I want you to publish your page using Quarto:

1. Create a Quarto website project locally
2. Render the site (Quarto creates a **_site/** folder)
3. Configure Quarto to output to **docs/**
4. Push to GitHub and enable Pages from the **docs/** folder

This will help you get comfortable Quarto, which we'll use for problem sets.

Research

What are you doing here?

Introductions

- Name
- Program/year
- Research interests
- Specific project (if applicable)
- One thing you like to do in your free time

This class

- The goal is translating theory and methods into actual research projects.
- Develop insight into causal thinking and provide tools for current and future research.
- Problem sets will help you practice.
- Presentations show you have done the readings and understand the assignments (somewhat AI proof).
- Exam will focus on intuition - critical for developing research questions and communicating methods/results.
- This works better with participation.

This class

- The material and tools are pivotal for **a lot** of what you will do in the future.
- Take responsibility for your education and career.
- Commit time to *understand* what you are doing.
- Be proactive and curious - syllabus is flexible.

General grad school thoughts

- You are the primarily one responsible for your success.
 - Work hard.
 - Think about how to utilize available resources.
- Grad school goes by quick
 - Good quote: *"I don't know what grad students do in their third year, but I know it takes them a year to do it"* – unattributed
- How can this class help you get to where you want to go?

Resource from physics

General grad school thoughts

Academic market

- You need papers
 - Ideally one is R&R or published when on the job market
 - Need to show independent work
 - Want to display initiative and a future pipeline

Non-academic market

- You want to have marketable skills (like R/python/causal inference)
- Think about what type of job you want (industry/government/etc.)

Course Material

Books

Causal Inference/Identification

- Mostly Harmless Econometrics *Angrist and Pischke, 2008*
- Causal Inference: The Mixtape *Cunningham 2020* **Link**
- The Effect *Huntington-Klein 2021* **Link** (last two available online)

Formal Theory

- Econometric Analysis *Greene, 2018*
- Microeconometrics: Methods and Applications *Cameron and Trivedi, 2005*

Papers

Overview papers (JEL/JEP)

- Abadie (2021) - Synthetic Control
- Roth et al. (2022) - Difference-in-difference
- Athey & Imbens (2019) - Machine learning

Methods

- Abadie, Diamond & Hainmueller (2010) - Synthetic Control
- Callaway & Sant'Anna (2020) - Difference-in-difference
- Athey & Imbens (2016) - Machine learning

Methods and applications

- Bueno & Valenta (2019) - Synthetic Control
- Marcus & Sant'Anna (2020) - Difference-in-difference
- Burlig, Knittel, Rapson, Reguant & Wolfram, (2019) - Machine learning

Our class

Mini-syllabus

- Attend/participate. *Spend time* on assigned readings — especially papers. This will include presentations on the papers. 20%
- Problem sets: 6-9 problem sets of replication and simulation in R. This will also include presentations of the assignments. 40%
- Final exam. This is written in class. 40%

Note: This class is under (constant) development.

Econometrics

What is special about econometrics relative to statistics or data science?

Causality

Causal research

Motivation

First, we believe that empirical research is most valuable when it uses data to answer specific causal questions, *as if* in a randomized clinical trial. This view shapes our approach to most research questions. In the absence of a real experiment, we look for well-controlled comparisons and/or natural quasi-experiments. Of course, some quasi-experimental research designs are more convincing than others, but the econometric methods used in these studies are almost always fairly simple.

Mostly Harmless Econometrics, p. xii (color added)

1. This ideology inherently compares research designs to “gold-standard” of RCTs
2. The methods are usually (relatively) straightforward (after training).

Angrist and Pischke's FAQs¹

1. What is the causal relationship of interest?
2. How would an ideal experiment capture this causal effect of interest?
3. What is your identification strategy?
4. What is your mode of inference?

¹ See *MHE*, chapter 1.

Note: Other questions also matter for developing quality research, e.g.,¹

- Why is your question important/interesting?
- Why is the current literature lacking or nonexistent?
- How do you propose to advance the literature?

¹ Credit for these questions goes to **Ed Rubin** via **Reed Walker**.

FAQ₁: What is the causal relationship of interest?

- Descriptive exercises can be **very interesting and important**, but in modern applied econometrics, causality is paramount.
- **Why?**
- Causal relationships directly test theories of how the world works.
- Causal relationships provide us with **counterfactuals**—how the world would have looked (or will look) under different sets of policies/circumstances.
- If you can't clearly and succinctly name the causal relationship of interest, then you may not actually have a research project.

FAQ₁: What is the causal relationship of interest?

Some classic and new examples...

Labor and Education

How does an additional year of schooling affect wages?

Urban

How do drivers respond to increased road capacity?

Environment

How much does improved air quality increase housing values?

Health

What is the effect of exposure to lead on test scores?

FAQ₂: What is the ideal experiment for this setting?

Describing the *ideal experiment* helps us formulate

- the exact causal relationship(s)
- the dimensions we want to manipulate
- the factors we need to hold constant

NOTE: These *ideal experiments* are generally hypothetical and often impossible. However, if you can't describe the ideal research design, it will be hard to come up with reasonable data and research designs in real life.

Angrist and Pischke call questions without ideal experiments *fundamentally unanswerable questions* (FUQs).

Experimental Ideal

Examples of potentially answerable questions...

- **The effect of education on wages:** Randomize scholarships or incentives to remain in school.
- **Urban:** Randomly allocate road infrastructure funding.
- **Environment:** Randomly take power stations or industrial plants offline.
- **Health:** Randomly test and improve lead fixtures in homes and schools.

FUQs

Examples of challenging questions (potentially unanswerable?)...¹

- How does gender affect eventual career paths?
- What role does race play in one's wages?

¹ Courtesy of **Ed Rubin**

FUQs

- Sometimes even simple-sounding policy questions turn out to be fundamentally unanswerable.

Example of a fundamentally unanswerable question: > Do children perform better by starting school at an older age?

Proposed ideal experiment

1. Randomize kids to start 1st grade at age 6 or 7.
2. Compare 2nd grade test scores.

Problem: Kids who start later are older in 2nd grade. Older kids do better. Do we want the effect of starting later or just being older?

FUQs

Example of a fundamentally unanswerable question: > Do children perform better by starting school at an older age?

Proposed ideal experiment_{2.0}

1. Randomize kids to start 1st grade at age 6 or 7.
2. Control for age. Compare test scores when kids are age 8.

Problem_{2.0} The two groups will have been in school for different numbers of years (1 vs. 2). More school should mean better scores.

FUQ

Example of a fundamentally unanswerable question:

Do children perform better by starting school at an older age?

Central problem: Mechanical links between ages and time in school.

$$(\text{Start Age}) = (\text{Current Age}) - (\text{Time in School})$$

No experiment can separate these effects (for school-age children).

FAQ₃: What's your identification strategy?

This question¹ describes how you plan to recover/observe *as good as random* assignment of your variable of interest (approximating your ideal experiment) in real life.

¹ You will hear this question asked *a lot*.

Examples

- Borders of implementation for daylight savings time
-**Energy consumption**
- License plates and lucky numbers in Beijing
-**Air quality; Driving; Congestion Costs**
- Thresholds or notches in policy eligibility
-**Pensions eligibility; Capital subsidies; R&D subsidies**
- Random judge assignments
-**Football games; Politics; Climate**

FAQ₃: What's your identification strategy?

The term “identification strategy” goes back to Angrist and Krueger (1991).

The variety of evidence presented in Section I establishes that compulsory schooling laws increase educational attainment for those covered by the laws. In Section II we consider whether students who attend school longer because of compulsory schooling receive higher earnings as a result of their increased schooling. Two-stage least squares (TSLS) estimates are used in which the source of **identification** is variation in education that results **solely from differences in season of birth-which, in turn, results from the effect of compulsory schooling laws**. The results suggest that men who are forced to attend school by compulsory schooling laws **earn higher wages as a result of their increased schooling**.

FAQ₄: What is your mode of inference?

- Historically, inference—standard errors, confidence intervals, hypothesis tests, *etc.*—has received much less attention than point estimates. It's becoming more important (more than an afterthought).
- Which **population** does your sample represent?
- How much **noise** (error) exists in your estimator (and estimates)?
- How much **variation** do you actually have in your variable of interest?
- Without careful inference, we don't know the difference between
 - $21\% \pm 2.3\%$
 - $21\% \pm 20.3\%$

Reading

Next Week

Required

- Susan Athey and Guido W Imbens (2017). “The state of applied econometrics: Causality and policy evaluation”. *Journal of Economic Perspectives* 31.2, pp. 3:32

Optional

- Guido W Imbens and Jeffrey M Wooldridge (2009). “Recent developments in the econometrics of program evaluation”. *Journal of economic literature* 47.1, pp. 5:86
- Arthur Lewbel (2019). “The identification zoo: Meanings of identification in econometric”. *Journal of Economic Literature* 57.4, pp. 835:903